Intro on Aishi" # Aishi – Academic Background, Research Experience, and Career Aspirations

Aishi is a third-year Bachelor of Technology (B.Tech) student pursuing Computer Science and Technology at Maharaja Agrasen Institute of Technology (MAIT), affiliated with Guru Gobind Singh Indraprastha University (GGSIPU), New Delhi. She belongs to the graduating batch of 2028 and has maintained a strong academic record with a CGPA of approximately 9.1 out of 10. Her primary academic interests include Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, Natural Language Processing (NLP), and applied AI research. She is particularly interested in developing intelligent systems that solve real-world problems and create a meaningful impact beyond traditional computer science applications.

In 2025, Aishi worked as a Science and Technology Intern at NITI Aayog, the Government of India's public policy think tank. During this internship, she contributed to a project called InfraVision-AI, which focused on identifying healthcare infrastructure using high-resolution satellite imagery. She worked with YOLOv8, Roboflow, and computer vision techniques to develop an object detection pipeline for identifying healthcare facilities across India's 112 Aspirational Districts. The project aimed to support data-driven decision-making and improve the monitoring of public healthcare infrastructure. This experience strengthened her understanding of dataset preparation, model training, evaluation, and the practical challenges of deploying machine learning solutions in government-related applications.

In April 2026, Aishi joined the Indian Institute of Technology Delhi (IIT Delhi) as a Machine Learning Research Intern, working under Professor Chetan Arora in collaboration with Larsen & Toubro (L&T). Her research focused on domain-robust object detection, particularly the challenge of selecting suitable object detection models for images originating from different visual domains. She worked on a research project titled "VLM-Guided Expert Routing for Domain-Robust Object Detection," which explored a training-free expert-routing approach. The system used visual embeddings from DINOv3 to compare incoming images with reference representations and identify the most appropriate specialized object detection expert. The approach achieved approximately 97.9% expert-selection accuracy in experimental evaluations. Her research involved studying vision-language models, experimenting with multiple object detection architectures, evaluating model performance, and contributing to the preparation of an academic research paper.

Beyond her internships, Aishi actively develops her programming and problem-solving skills. She has solved more than 350 problems on LeetCode and primarily uses Python and C++ for algorithmic problem-solving. Her technical experience includes PyTorch, scikit-learn, Hugging Face, Keras, NumPy, OpenCV-related computer vision workflows, and various machine learning frameworks. She is also familiar with backend development, REST APIs, SQL databases, Git, Linux, and web development technologies such as JavaScript, React, Flask, and FastAPI. Recently, she has been exploring Retrieval-Augmented Generation (RAG), vector databases, embedding models, and large language models to understand how modern AI applications retrieve and generate information.

Aishi has also participated actively in hackathons and technical competitions. She has been a finalist in five hackathons, won two hackathons, and participated as a finalist in Smart India Hackathon 2025. She was selected as a SheFi 2025 Cohort Scholar and received a scholarship worth approximately $800. Additionally, she has served as the Technology Head of TechXtract, the official Computer Science and Engineering technical society at her college. Through these experiences, she has developed skills in teamwork, technical leadership, project management, and communicating complex technical ideas.

Apart from her technical interests, Aishi enjoys exploring literature, English, music, and subjects outside conventional engineering. She values interdisciplinary learning and is interested in how artificial intelligence can interact with fields such as education, public policy, and the social sciences. Although much of her academic experience has involved computer vision, she hopes to expand into broader machine learning research, including language models, intelligent agents, and general-purpose AI systems.

Looking ahead, Aishi plans to pursue a master's degree in Artificial Intelligence or Machine Learning at an international university. Her long-term career aspiration is to become an AI researcher, research engineer, or applied scientist, working on technologies that combine scientific innovation with practical societal impact. She is particularly motivated by research that can benefit people outside the immediate AI and computer science communities.